

Case on using Statistical tools for financial decision making (For class discussion)

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General Insurance Corporation of India

I. Case:

The management of the General Insurance Corporation of India asked the marketing research manager to conduct a survey to determine consumer interest in a new auto insurance concept then under consideration. The concept involved coverage for costs incurred as a result of a car breaking down while on a trip.

The manager designed and conducted a small telephone survey involving a sample of 60 respondents. Half the sample was drawn randomly from the company's present customer list, and half was drawn by means of random-digit dialing. In each interview the following statement describing the proposed insurance coverage was read to the respondent over the telephone:

"An insurance company is considering providing a policy that would pay toward expenses incurred when your car breaks down while you are driving more that 350 kms from home. It pays up to Rs.20000 toward such items as towing, repair costs, other transportation, rooms, and meals. The policy will cost Rs.500 a year.

Following the reading of this statement, the respondent was asked to state his or her degree of interest in the policy on a 5-point scale ranging from 1 (definitely not interested) to 5 (very much interested). The respondent was then asked to supply a small amount of background data regarding: Age, Marital status, Number of cars owned, Average age of car(s), Number of trips taken by automobile (for any purpose whatsoever) over the past year that exceeded 300 miles on a round-trip basis.

The data obtained from the survey are shown in Table I.

Table I. Telephone survey data

Respon -dent	Concept Rating	Current Insurance Supplier	Age	Marital Status	Number of Cars	Average Age of Car(s)	Number of Trips
1	4	General Ins.	42	M	1	0.5	3
2	3	G	39	M	1	1.5	1
3	5	G	47	M	1	1	4
4	2	G	24	S	3	1	2
5	4	G	43	M	2	1.5	4
6	5	G	62	M	1	0.5	6
7	1	G	27	M	1	2	3
8	5	G	55	M	2	0.5	4
9	4	G	42	S	1	2	2
10	3	G	36	S	1	2	2
11	4	G	39	M	3	1.5	5
12	2	G	24	S	4	2	0

13	5	G	58	M	1	2	6
14	4	G	43	M	1	0.4	2
15	1	G	23	S	2	2.5	0
16	5	G	59	M	1	0.5	7
17	4	G	43	M	3	1.5	3
18	3	G	36	S	1	2	0
19	3	G	47	M	1	2	1
20	4	G	42	M	1	1.5	4
21	4	G	47	M	4	1.5	4
22	4	G	38	M	2	1	3
23	5	G	37	M	2	0.8	8
24	3	G	39	S	1	2	0
25	4	G	51	M	1	1	2
26	4	G	47	M	2	1.5	1
27	5	G	51	M	1	2	6
28	1	G	30	S	1	2	0
29	1	G	28	S	2	4.5	2
30	3	G	42	M	2	3.5	1
31	2	Other	32	M	3	3	0
32	4	O	29	M	1	2	3
33	2	O	32	M	1	1	0
34	1	O	37	M	1	1	0
35	3	O	24	S	4	2.5	1
36	2	O	41	M	1	2	3
37	3	O	23	M	1	2	0
38	1	O	34	M	5	3	1
39	2	O	38	M	2	2	0
40	4	O	47	M	2	1	5
41	5	O	24	M	1	0.5	0
42	3	O	32	M	1	1.5	0
43	1	O	22	S	1	3	1
44	2	O	27	S	1	2.5	0
45	2	O	29	M	3	2.5	0
46	4	O	43	M	2	1	2
47	5	O	48	S	1	0.5	3
48	3	O	36	M	1	1.5	0
49	4	O	42	M	3	1.5	2
50	2	O	26	S	2	2	2
51	2	O	29	S	1	2.5	1
52	1	O	23	S	1	3	0
53	3	O	34	M	1	1.5	0
54	4	O	37	S	1	1.5	2
55	2	O	24	S	2	3	0
56	3	O	32	M	2	2	1
57	5	O	44	M	1	0.5	7
58	1	O	28	M	1	2.5	0
59	1	O	22	S	2	3	1
60	2	O	26	S	1	2	1

What is the wisdom of offering the policy? Its target market? Trading strategy?

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Issues for analysis:

1. How does degree of interest in the new concept appear to be related to:
 - a. Whether the respondent is a current General Insurance customer or not?
 - b. Respondent age?
 - c. Number of automobile trips?
2. What is the representative age and typical number of automobile trips? To what extent can this representative value be used for decision making?
3. Are there any outliers? What do we do if there are outliers?
4. Are all the variables normally distributed?
5. Is there any association between concept rating and the company from which insurance has been taken by the customer?
6. Is the association different for different age groups and concept rating?
7. What variables influence the concept rating? What is the significance of the variables in influencing the concept rating?
8. Divide the sample into four groups: General Insurance, single; General Insurance, married; other company, single; and other company, married. Is there any significant difference in the concept rating and the four groups of customers?
9. Divide the sample into two groups: those rating the concept "4" or "5" versus those rating it "1," "2," or "3." What characteristics are different for the two groups?
10. Having completed the various analyses, what are your recommendations regarding:
 - a. The wisdom of offering the policy?
 - b. Its target market?
 - c. Future research to take before actual introduction of the policy?